



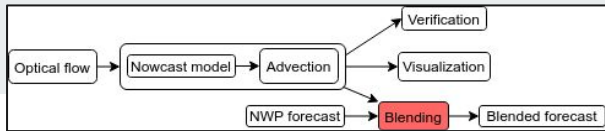
Hands-on session “Blending with NWP”

Exercises



Session overview

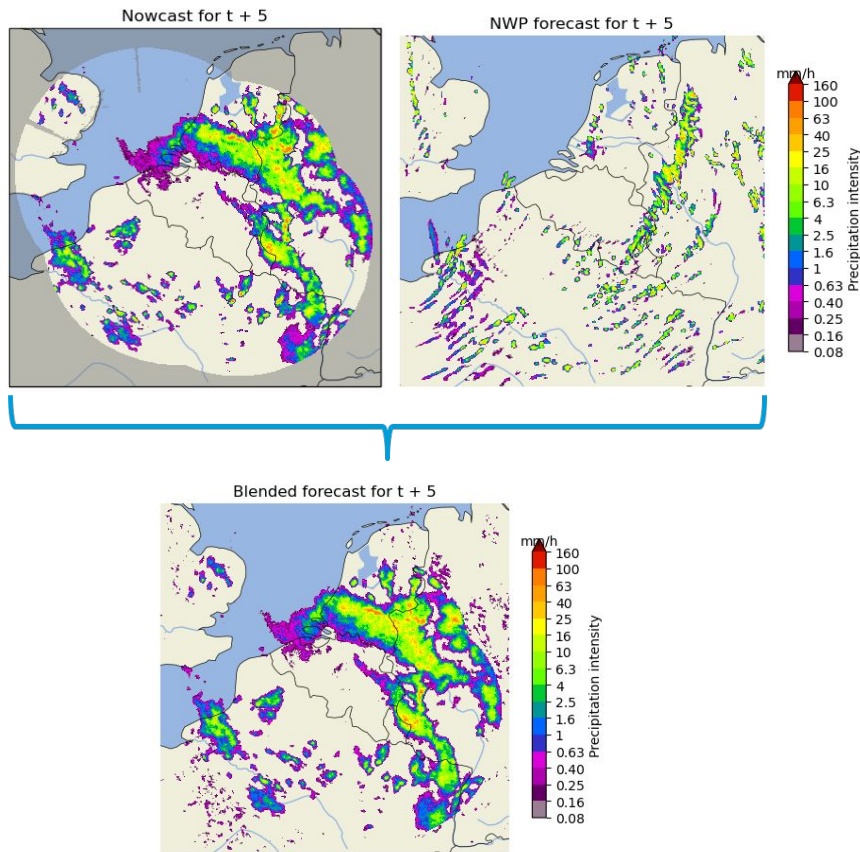
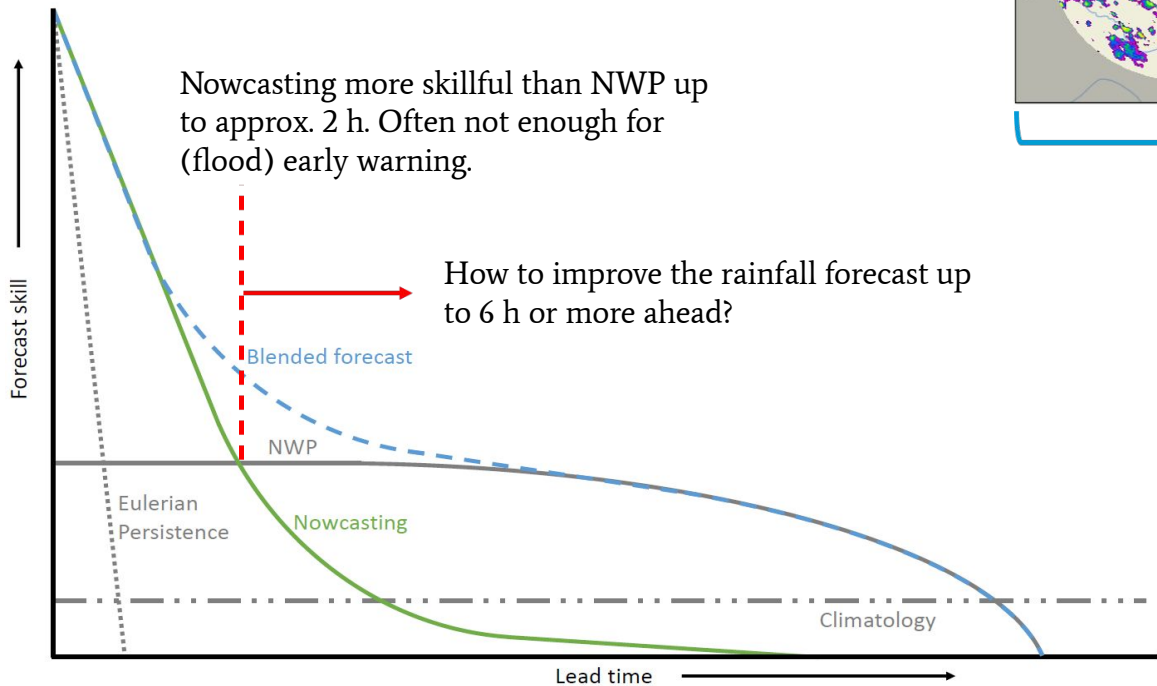
Time	Content	Speaker	Duration
09:00	Welcome / Course overview	Ruben	20'
09:20	Lecture: "Pysteps in operations"	Ruben, Lesley	40'
10:00	Coffee break		30'
10:30	Hands-on session "Starting with pysteps"	Jenna, Miguel, Mats	120'
12:30	Lunch break		60'
13:30	Hands-on session "Blending with NWP"	Mats, Lesley, Ruben	120'
15:30	Coffee break		30'
16:00	Lecture: "AI for nowcasting"	Lesley, Jenna, Ruben	45'
16:45	Discussion	All	15'
17:00	End of the short course		



Blending with NWP

Nowcasting more skillful than NWP up to approx. 2 h. Often not enough for (flood) early warning.

How to improve the rainfall forecast up to 6 h or more ahead?

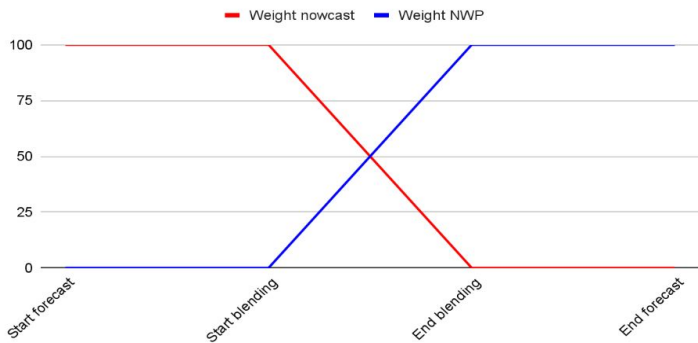


Blending with NWP: methods in pysteps

1. Linear blending

- Fixed start and end point of blending procedure
- Weights go linearly from 1 to 0 and 0 to 1.

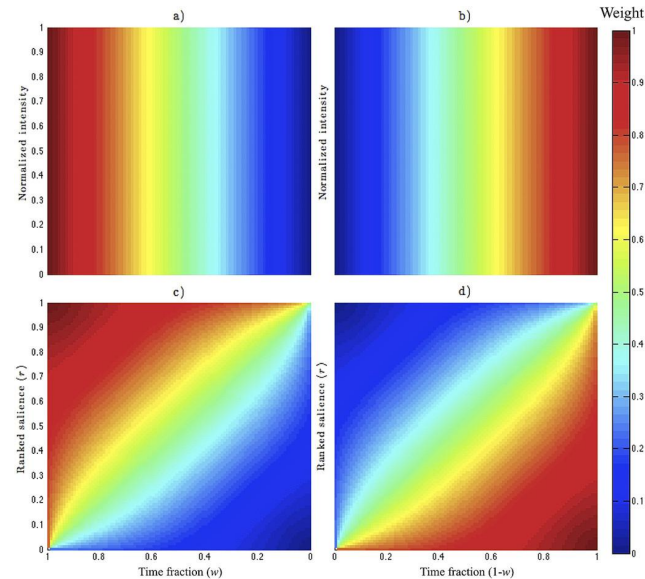
Linear blending weights



2. Saliency-based blending

- Similar to linear blending, but:
- Preserves pixel intensities over time if they are strong enough according to their ranked salience.

3. STEPS blending (see next slide)



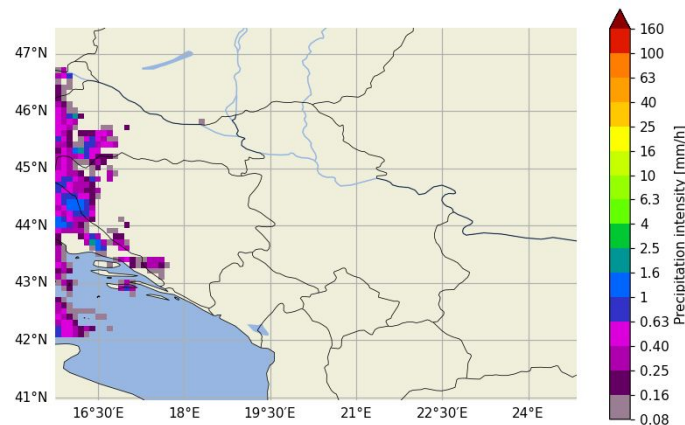
Exercise 5: Blending with NWP

For this exercise, we will use both the Serbian **radar** data and output from a numerical weather prediction (**NWP**) model.

We will use hourly output from **IFS**, the global weather model of the European Centre for Medium-range Weather Forecasts (ECMWF).

ECMWF IFS control run, initialised 2017-08-12 00 UTC

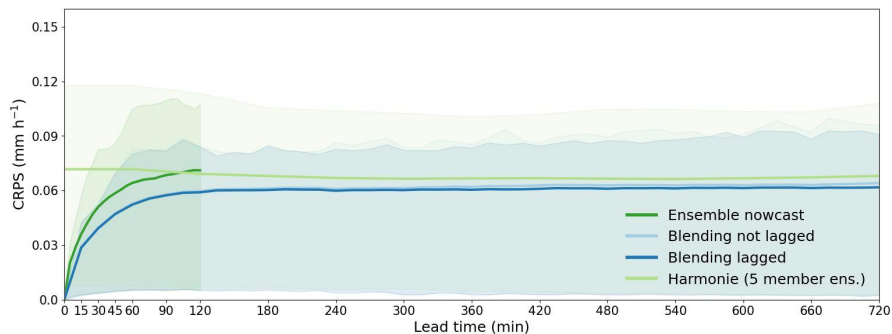
2017-08-12 06:00 - 07:00 UTC



Exercise 5: Blending with NWP

The preliminaries for this exercise are in *helper_blending.ipynb*, feel free to have a look at it as a starting point for this exercise.

Try out a linearly, saliency-based and STEPS blended nowcast. Feel free to play around with it!



Linear blending - adjustable settings:

- Start and end time blending
- Nowcasting method
- Probabilistic forecast

Saliency-based blending: better preservation of high-intensity rainfall cells during blending

STEPS blending - other adjustable settings:

- Weights determination method
- Skill window size
- Option to store decomposed NWP forecast
- Blend ensemble nowcast with ensemble NWP